Math 110 — Week 3

Solutions

HW3 P1.

Idea. A formula with dim null plus another dimension is rank–nullity for some map. The left side is $\dim T^{-1}(U)$, so the map should have domain $T^{-1}(U)$: restrict T to it.

1. (a) $T0 = 0 \in U$. If $Tv, Tw \in U$ then $T(\lambda v + w) = \lambda Tv + Tw \in U$.
2. (b) Let $S\colon T^{-1}(U) \to W$, $Sv = Tv$. Its domain is finite-dimensional (a subspace of V).
3. $\operatorname{null} S = \operatorname{null} T$. If $Tv = 0$, then $Tv = 0 \in U$, so $v \in T^{-1}(U)$. Hence $\operatorname{null} T \subseteq T^{-1}(U)$, and $\operatorname{null} S = \{v \in T^{-1}(U) : Tv = 0\} = \operatorname{null} T$.
4. $\operatorname{range} S = U \cap \operatorname{range} T$: $Sv = Tv$ lies in U and in $\operatorname{range} T$. Conversely if $w \in U \cap \operatorname{range} T$, then $w = Tv$ with $Tv \in U$, so $v \in T^{-1}(U)$ and $w = Sv$.
5. Rank–nullity for S: $\dim T^{-1}(U) = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T)$.
6. (c) The space is $D^{-1}(U)$ with $U = \operatorname{span}(1, x^2)$. $\operatorname{null} D = \operatorname{span}(1)$, dimension 1. $\operatorname{range} D = \mathcal{P}_3(\mathbb{R}) \supseteq U$, so $U \cap \operatorname{range} D = U$, dimension 2. Answer: $1 + 2 = 3$. (A basis: $1,\ x,\ x^3$.)

HW3 P2.

Idea. N on $\mathbb{R}^2$ has $\operatorname{null} N = \operatorname{range} N$; use two copies. For (b), rank–nullity makes the dimension even.

1. (a) Define T on the standard basis (Prop. 21): $Te_1 = 0$, $Te_2 = 0$, $Te_3 = e_1$, $Te_4 = e_2$, that is,

$$T(x_1, x_2, x_3, x_4) = (x_3, x_4, 0, 0).$$

   Then $\operatorname{range} T = \operatorname{span}(e_1, e_2)$ and $\operatorname{null} T = \{x_3 = x_4 = 0\} = \operatorname{span}(e_1, e_2)$.
2. (b) If $\operatorname{null} T = \operatorname{range} T$, then $5 = \dim \operatorname{null} T + \dim \operatorname{range} T = 2 \dim \operatorname{null} T$, which is impossible since 5 is odd.

HW3 P3.

Idea. The two conditions are the two coordinates of a linear map to $\mathbb{R}^2$; if it is surjective the null space has dimension $5 - 2$.

1. $S(p) = (p(1), p'(1))$ is linear (evaluation and differentiation are), and $\operatorname{null} S = U$.
2. S is surjective: $S(1) = (1, 0)$ and $S(x - 1) = (0, 1)$ span $\mathbb{R}^2$.
3. $\dim U = \dim \mathcal{P}_4(\mathbb{R}) - \dim \operatorname{range} S = 5 - 2 = 3$.
4. Homework 2 found the basis $(x-1)^2, (x-1)^3, (x-1)^4$ directly. Rank–nullity gives the dimension without finding a basis.

HW3 P4.

Idea. A linear map sends a relation to a relation with the same coefficients.

1. (a) If $\sum \lambda_j v_j = 0$ with not all $\lambda_j = 0$, then

$$\sum \lambda_j T v_j = T(\sum \lambda_j v_j) = T0 = 0$$

   with the same coefficients.
2. (b) $T = 0\colon \mathbb{R} \to \mathbb{R}$ sends the independent list 1 to the dependent list 0. (Or $P(x, y) = (0, y)$ sends e_1, e_2 to $0, e_2$.)

HW3 P5.

Idea. Apply T to each monomial; the results are the columns.

1. (a) $T(1) = 1$. $T(x) = x^2 + x$. $T(x^2) = 2x^3 + x^2$. $T(x^3) = 3x^4 + x^3$. So
$$\mathcal{M}(T) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$
2. (b) For $p = a + bx + cx^2 + dx^3$,
$$Tp = a + bx + (b + c)x^2 + (2c + d)x^3 + 3dx^4.$$
If $Tp = 0$: $a = 0$, $b = 0$, then $c = 0$, then $d = 0$. So $\operatorname{null} T = \{0\}$, dimension 0.
3. By rank–nullity $\dim \operatorname{range} T = 4$, and $\operatorname{range} T = \operatorname{span}(1,\ x + x^2,\ x^2 + 2x^3,\ x^3 + 3x^4)$.
4. (c) $\operatorname{range} T$ is spanned by the images of the basis vectors $1, x, x^2, x^3$. Column j is the coordinate vector of the j-th image. So in coordinates the range is the span of the columns.

HW3 P6.

Idea. Use the second equation to eliminate z; in $\mathbb{F}_2$, $x + x = 0$.

1. $x + z = 0$ gives $z = -x = x$.
2. Then $x + y + z + w = 2x + y + w = y + w = 1$, so $w = y + 1$.
3. x, y are free. Solutions (x, y, z, w): $(0, 0, 0, 1)$, $(0, 1, 0, 0)$, $(1, 0, 1, 1)$, $(1, 1, 1, 0)$.

HW3 P7.

Idea. Two solutions differ by an element of the null space, and a k-dimensional space over $\mathbb{F}_2$ has 2^k elements.

1. Fix one solution x_0. Then $Ax = b \iff A(x - x_0) = 0 \iff x \in x_0 + \operatorname{null} A$. So the number of solutions is the number of elements of $\operatorname{null} A$.
2. A is a map $\mathbb{F}_2^m \to \mathbb{F}_2^n$, so $\dim \operatorname{null} A = m - \operatorname{rank} A$ (rank–nullity).
3. A space of dimension k over $\mathbb{F}_2$ has 2^k elements (each of k coordinates is 0 or 1). Answer: $2^{m - \operatorname{rank} A}$.

R1.

Idea. Columns are the images of the vectors of B, written in $1, x, x^2$. Equal dimensions: injective is enough.

1. $T(1, 0, 0) = 1 + x^2$, $T(1, 1, 0) = 2 + x + x^2$, $T(1, 1, 1) = 2 + 2x + 2x^2$. So
$$\mathcal{M}(T) = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix}.$$
2. (b) If $T(a, b, c) = 0$: $a + b = 0$, $b + c = 0$, $a + c = 0$. Then $a = -b = c$ and $a + c = 2a = 0$, so $a = b = c = 0$. T is injective.
3. $\dim \mathbb{R}^3 = 3 = \dim \mathcal{P}_2(\mathbb{R})$, so T is also surjective (Prop. 25).

R2.

Idea. The formula of HW3 P1(b) needs $\operatorname{null} D$ and $U \cap \operatorname{range} D$; for a basis, add to a basis of $\operatorname{null} D$ one preimage of each basis vector of U.

1. $\operatorname{null} D = \operatorname{span}(1)$, of dimension 1. $\operatorname{range} D = \mathcal{P}_3(\mathbb{R})$ contains U, so $U \cap \operatorname{range} D = U$, of dimension 2.
2. By HW3 P1(b),
$$\dim D^{-1}(U) = \dim \operatorname{null} D + \dim(U \cap \operatorname{range} D) = 1 + 2 = 3.$$
3. $1 \in D^{-1}(U)$ since $D1 = 0$; $x \in D^{-1}(U)$ since $Dx = 1$; $x^3 \in D^{-1}(U)$ since $Dx^3 = 3x^2$. These three nonzero polynomials have different degrees $0, 1, 3$, so they are independent (staircase lists are independent, Week 2: list them by degree; each has a nonzero coefficient of a power where the earlier ones have 0), and three independent vectors in a space of dimension 3 form a basis.

R3.

Idea. Column j holds the coordinates of T applied to the j-th basis polynomial.

1. $T(1) = 0$, $T(x) = 1$, $T(x^2) = 2 + 2x$, $T(x^3) = 6x + 3x^2$.
2. Writing each image in $1, x, x^2$ and using the coordinate columns as columns,
$$\mathcal{M}(T) = \begin{bmatrix} 0 & 1 & 2 & 0 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

R4.

Idea. A map to $\mathbb{R}^2$ has rank at most 2, and rank–nullity then bounds the null space from below.

1. $\operatorname{range} T \subseteq \mathbb{R}^2$, so $\dim \operatorname{range} T \leq 2$.
2. Rank–nullity gives $\dim \operatorname{null} T = 4 - \dim \operatorname{range} T \geq 2$.
3. The given set is $\operatorname{span}\big((1,1,1,1)\big)$, of dimension 1. So no such T exists.

R5.

Idea. The rank is at most the dimension of the domain; the images of $1, x, x^2$ give the matrix, the range, and (by their independence) the null space.

1. (i) $\dim \operatorname{range} S \leq \dim \mathcal{P}_2(\mathbb{R}) = 3 < 4 = \dim \mathcal{P}_3(\mathbb{R})$, so S is not surjective.
2. (iii) $S(1) = x$, $S(x) = \frac{1}{2}x^2 - 1$, $S(x^2) = \frac{1}{3}x^3 - 2x$. Their coordinates in $1, x, x^2, x^3$ are the columns:
$$\mathcal{M}(S) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -2 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}.$$
3. (ii) The three images are nonzero polynomials of degrees $1, 2, 3$, so they are independent (staircase lists are independent, Week 2); hence $\dim \operatorname{range} S = 3$, $\operatorname{range} S = \operatorname{span}\big(x, \frac{1}{2}x^2 - 1, \frac{1}{3}x^3 - 2x\big)$, and by rank–nullity $\dim \operatorname{null} S = 0$, so $\operatorname{null} S = \{0\}$.

R6.

Idea. A projection splits the plane into the line it kills and the line it keeps.

1. Take $P(x, y) = (0, y)$. Then $\ker P = \operatorname{span}(e_1)$ and $\operatorname{range} P = \operatorname{span}(e_2)$.
2. $\operatorname{span}(e_1) \cap \operatorname{span}(e_2) = \{0\}$. Every $(x, y) = (x, 0) + (0, y)$ lies in $\ker P + \operatorname{range} P$, and $\ker P + \operatorname{range} P \subseteq \mathbb{R}^2$, so $\ker P + \operatorname{range} P = \mathbb{R}^2$. Hence $\mathbb{R}^2 = \ker P \oplus \operatorname{range} P$ (Def. 6).